

MCU Capability Analyzer

EFFORT 4-6 hours	HARDWARE Optional	AI TOOLS Allowed	OUTPUT CLI + JSON
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Purpose: Create a lightweight analyzer that converts official MCU, board, and SDK information into a traceable capability report.

1. Objective

Choose any publicly available MCU development board and build a small command-line tool that creates a structured capability report for it.

Physical hardware is not required. AI tools may be used freely, but all important technical information must be verified against official sources.

2. Inputs

The tool should accept:

- MCU or development board name
- Official SDK path or repository
- Optional datasheet, board manual, or documentation references

Example command:

```
analyze --board <board-name> --sdk <sdk-path>
```

3. Capability Report

Generate a board_capabilities.json file containing a structured summary similar to:

```
{
  "vendor": "", "board": "", "mcu": "",
  "cpu_core": "", "flash_kb": 0, "ram_kb": 0,
  "sdk": {"name": "", "version": ""},
  "board_resources": {
    "led": {"pin": "", "active_level": ""},
    "debug_uart": {
      "instance": "", "tx_pin": "", "rx_pin": ""
    }
  },
  "peripherals": {
    "gpio": true, "uart": true, "adc": true,
    "timer": true, "spi": true, "i2c": true,
    "can": false, "usb": false, "ethernet": false
  }
}
```

Source Traceability: Each important field must include a reference to the official source from which it was identified.

4. SDK Example Search

Identify the closest official SDK example for:

- LED blinking
- UART communication
- ADC reading
- Timer-based periodic execution

Example output:

```
{
  "requirement": "periodic ADC reading over UART",
  "recommended_examples": [
    {
      "path": "examples/adc/basic",
      "reason": "Contains ADC initialization and polling"
    },
    {
      "path": "examples/uart/echo",
      "reason": "Contains UART initialization and transmission"
    }
  ]
}
```

5. Validation Task

Use the generated capability report to answer:

Question: Can this board support UART logging, periodic ADC sampling, and native Ethernet?

Classify each feature as:

- Supported
- Supported by the MCU but not available on the board
- Requires an external component
- Unsupported
- Not enough verified information

Important: Do not invent peripherals, APIs, pin mappings, or board capabilities. Unknown information must remain explicitly unknown.

6. Deliverables

- Source code
- README with setup and execution instructions
- board_capabilities.json
- SDK example recommendations
- Capability validation report
- Short note explaining how AI was used and verified
- Demonstration video of no more than five minutes

7. Constraints

- Expected effort: 4-6 focused hours.
- A graphical interface is not required.
- Physical hardware is optional.
- Use only official SDKs and documentation for technical claims.
- Pin mappings and capabilities must be traceable.
- Unknown information must be marked as unknown, not assumed.

8. Evaluation Criteria

Evaluation Area	What We Will Look For
MCU and board accuracy	The report correctly distinguishes MCU capabilities from board-level resources.
SDK analysis	Relevant official examples are located and recommended for clear reasons.
Source traceability	Important claims can be traced to reliable official documentation.
Capability classification	Supported, unavailable, external, and unsupported cases are classified correctly.
Unknown handling	Missing evidence is reported honestly rather than guessed.
Reusability	The design can be extended to analyze another MCU or board with limited changes.

Submit the repository, generated outputs, and demonstration link together in a single response.